GENERATIVE AI

Diffusion Models Visual Guide

How image generators work:
noising, denoising, U-Net, sampling, conditioning.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/5

Forward: Adding Noise

Destroy the image on a fixed schedule.

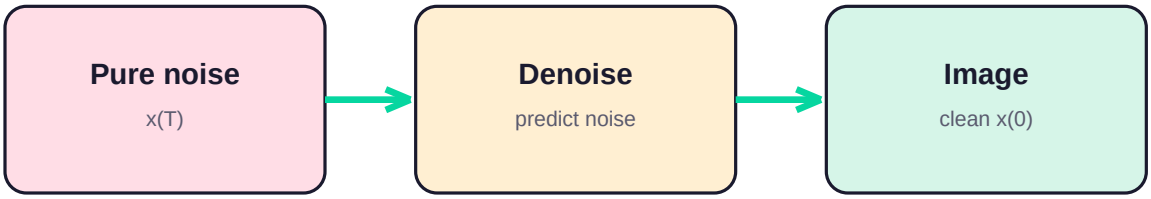
Each step adds a little Gaussian noise. After T steps the image is pure static — a known distribution.

The forward process needs no learning — it is a fixed recipe. All the intelligence lives in the reverse process.

STEP 2/5

Reverse: Denoising

Learn to remove the noise, one step at a time.



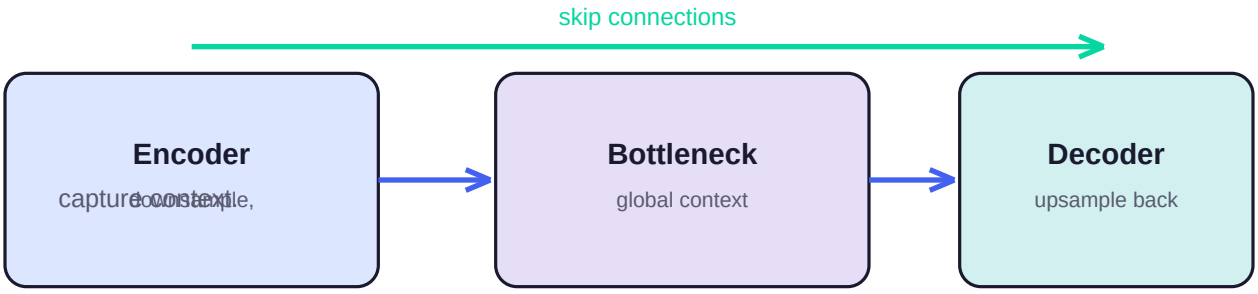
The model predicts the noise added at step t, subtracts it, and repeats — walking noise back into an image.

Training: take a clean image, add known noise, ask the model to predict that noise. Simple MSE loss.

STEP 3/5

The U-Net

The denoiser's architecture.

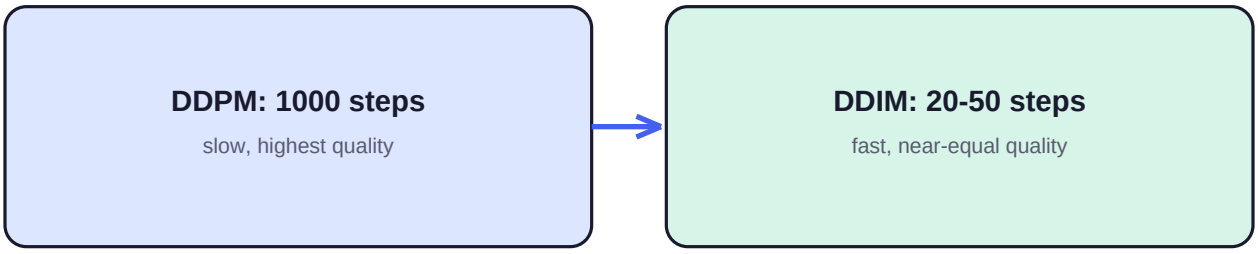


Skip connections carry fine detail past the bottleneck — sharp edges survive the round trip.

STEP 4/5

Sampling: Steps vs Speed

Fewer steps, faster images — with tradeoffs.



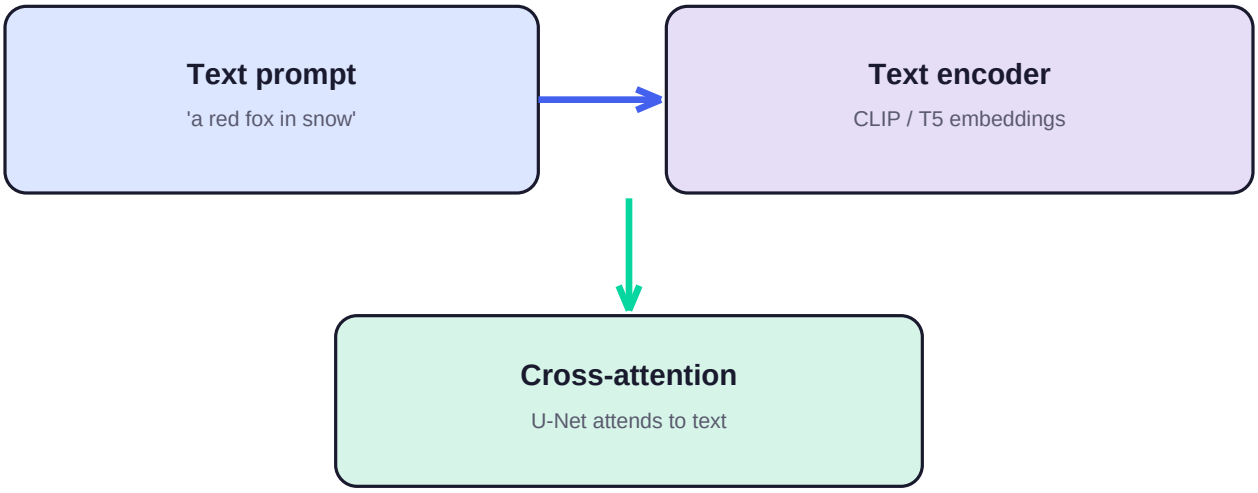
DDIM skips steps deterministically. Distilled models (few-step) push generation to 1-8 steps.

The sampler is swappable after training — upgrade speed without retraining the model.

STEP 5/5

Text Conditioning

Steering generation with words.



Classifier-free guidance: blend conditional and unconditional predictions — higher guidance = stronger prompt adherence.

STEP ✓I✓

Key Takeaways

Diffusion Models — Remember

- Forward process adds noise on a fixed schedule; no learning needed there.
- The model learns reverse denoising: predict the noise, subtract, repeat.
- U-Net with skip connections is the classic denoiser architecture.
- DDIM and distilled samplers cut 1000 steps to dozens with little quality loss.
- Text encoders + cross-attention + classifier-free guidance steer generation.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev